Rack-Level Micro-Climate & Growth Optimizer for Oyster Mushroom Farms

R26-ISE-002

Project Proposal Report

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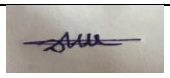
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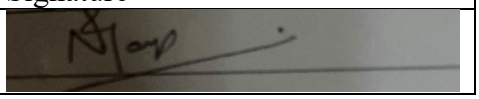
March 2026

DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

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H.P.D.M.Chandrasiri	**IT22180148**	

The above candidate is conducting research for their undergraduate Dissertation under my supervision.

Name	Date	Signature
Ms.N.Gamage	13/3/2026	

ABSTRACT

Oyster mushroom cultivation requires careful monitoring of growth stages to achieve the best harvest time and higher yield. However, traditional manual observation methods are time-consuming and can lead to mistakes, especially for new farmers who do not have enough experience. There is currently no simple automated system that helps small-scale or beginner farmers identify mushroom growth stages and predict harvest time. This project proposes a Biological Stage Detection System, a mobile and web-based solution that uses image processing and machine learning to analyze mushroom images captured by a camera. The system identifies the biological stage such as early growth, pinning, mature, or ready-to-harvest and estimates the remaining days until harvest. The results are shown through a user-friendly interface with color indicators and notifications to help farmers make better decisions. The system will be developed using React Native for the mobile application, Node.js and Express.js for backend services, MongoDB for data storage, and TensorFlow or PyTorch for the machine learning model, while temperature and humidity sensors will support environmental monitoring. This solution is expected to achieve around 90% accuracy in stage detection, predict harvest time within ±1 day of expert estimates, and help farmers improve monitoring efficiency, increase yield, and reduce losses.

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to all the individuals who supported me throughout the development of the Biological Stage Detection System for oyster mushrooms.First and foremost, I wish to thank my project supervisor and co-supervisor for their valuable guidance, advice, and encouragement throughout the research and development process. Their expertise helped me define the project scope, overcome challenges, and stay focused on the objectives.I am also deeply grateful to my core supervisor for their continuous support, critical insights, and mentorship, which were instrumental in shaping the technical aspects of the system and ensuring it met both practical and academic standards.

Finally, I extend my gratitude to all others who directly or indirectly contributed to this project. This work reflects the collective support, guidance, and encouragement I received, and I am sincerely thankful to everyone who helped make this project possible.

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LIST OF ABBREVIATIONS

RP	Research Project
FAO	Food and Agriculture Organization
CNN	Convolutional Neural Network
SDG	Sustainable Development Goals
IOT	Internet of Things
SDLC	Software Development Life Cycle
UI	User Interface
ML	Machine Learning
AI	Artificial Intelligence
DL	Deep Learning
API	Application Programming Interface
DB	Database
GPS	Global Positioning System
HTTP	Hypertext Transfer Protocol
IDE	Integrated Development Environment
RGB	Red Green Blue
UX	User Experience
OS	Operating System
KPI	Key Performance Indicator

1. INTRODUCTION

1.1 Background and Context

Mushroom cultivation has become an important agricultural activity around the world due to its high nutritional value and increasing demand in the food industry. Among different mushroom varieties, oyster mushrooms are widely cultivated because they grow quickly and require relatively simple environmental conditions. However, successful mushroom cultivation requires careful monitoring of the biological growth stages of mushrooms. These stages include early growth, pinning, mature stage, and ready-to-harvest stage. Identifying the correct stage is very important because harvesting mushrooms too early or too late can reduce their quality and market value. In most mushroom farms, farmers identify mushroom stages through manual observation based on their experience. This traditional method can sometimes lead to inaccurate decisions, especially for beginner farmers who may not be familiar with the different stages of mushroom growth.

This issue mainly affects small-scale mushroom farmers who depend on manual monitoring to manage their farms. A typical mushroom house contains multiple racks where mushrooms grow simultaneously, and mushrooms on each rack may develop at different rates. Because of this, farmers must constantly check each rack and remember the stage of mushrooms in different locations. Without proper monitoring tools, farmers may find it difficult to track the development of mushrooms in every rack. This can lead to several problems such as reduced mushroom quality, lower harvest yield, and increased time spent on manual observation. Therefore, improving the monitoring process is important to help farmers make better harvesting decisions and increase their productivity.

Globally, mushroom production has increased significantly due to the growing demand for healthy and sustainable food sources. According to reports from the Food and Agriculture Organization (FAO), global mushroom production has continued to rise over the past decade as mushrooms provide a good source of protein, vitamins, and minerals. In Sri Lanka, mushroom cultivation has also become popular as a small-scale agricultural business because it requires relatively low investment and limited space. Many local farmers cultivate oyster mushrooms in small mushroom houses using rack systems. However, most farmers still rely on traditional farming practices and manual observation methods without technological support for monitoring mushroom growth stages.

Several research studies have explored the use of image processing and machine learning techniques in agricultural monitoring systems. Image-based classification methods are commonly used to detect plant diseases, fruit ripeness, and crop growth stages by analyzing visual characteristics such as color, shape, and texture. Deep learning models such as Convolutional Neural Networks (CNNs) have shown high accuracy in identifying biological features from images. Some research has also applied image analysis to mushroom cultivation to detect mushroom size, cap development, and maturity level. Although these systems help farmers understand mushroom growth conditions, most of them mainly focus on general crop monitoring or simple maturity detection and do not provide practical management features for real mushroom farms.

Because of the limitations of existing systems, there is a need for a more practical solution that can support farmers in monitoring mushroom growth more effectively. The proposed system introduces an image-based biological stage detection component that allows farmers to capture images of mushrooms using a fixed camera attached to a portable device or sensor system. The captured images are sent to the system for analysis. The system then processes these images using image processing techniques and machine learning models to identify the biological stage of the oyster mushroom and estimate the remaining number of days until harvesting. In addition, the system includes a rack-wise monitoring feature that allows farmers to record and track the growth stage of mushrooms for each rack separately. This enables farmers to easily identify which racks are ready for harvesting and manage their mushroom farms in a more organized and efficient manner.

The proposed system also aligns with the United Nations Sustainable Development Goals (SDGs), particularly SDG 2: Zero Hunger, which focuses on improving agricultural productivity and promoting sustainable food production systems. By providing a technological solution that supports efficient mushroom cultivation, the system can help improve food production and support small-scale farmers. Furthermore, this research aligns with the Smart Agriculture and Artificial Intelligence research clusters, as it integrates modern technologies such as machine learning, image processing, sensor devices, and mobile applications to develop an intelligent agricultural monitoring system.

1.2 Literature Review

Several studies have explored the use of image processing and machine learning techniques in agriculture to monitor crop growth and support farmers in making better decisions. Image-based systems are commonly used to analyze plant characteristics such as size, color, texture, and shape, which helps researchers detect plant diseases, identify crop maturity levels, and understand growth stages. In particular, deep learning methods such as Convolutional Neural Networks (CNNs) have been widely used for plant image classification and agricultural monitoring because they can automatically extract visual features from images and classify them with high accuracy. In mushroom cultivation, several studies have proposed systems that analyze mushroom images to monitor growth and determine harvesting time. For example, some research has developed image-processing-based systems that analyze oyster mushroom images and identify harvest readiness by examining visual features such as color, texture, and size, achieving high accuracy levels. Other studies have also used computer vision techniques and object-detection models such as YOLO to detect mushroom growth stages and support farmers in deciding the correct harvesting time.[1],[3],[4],[5]

In addition, researchers have developed smart monitoring systems that combine sensor technologies, image processing, and machine learning for mushroom farming. These systems often use Internet of Things (IoT) devices together with CNN-based models to monitor environmental conditions, detect diseases, and estimate harvesting time for mushrooms. The results of these studies show that intelligent agricultural systems can improve farming efficiency and productivity. However, most existing solutions mainly focus on mushroom detection, species identification, environmental monitoring, or harvest readiness prediction. They do not provide rack-wise monitoring features that allow farmers to track the biological growth stage of mushrooms separately for each rack in a mushroom house. Since mushroom farms usually contain multiple racks with mushrooms growing at different stages, the lack of rack-level monitoring makes it difficult for farmers to manage harvesting effectively. Therefore, the proposed system aims to address this limitation by combining image-based biological stage detection with rack-wise monitoring, enabling farmers to easily track the growth stage of mushrooms in each rack and manage their farms more efficiently.[2],[6],[7],[8]

1.3 Research Gap

Current agricultural monitoring systems and mushroom farming technologies mainly focus on general crop monitoring, environmental condition tracking, or simple harvest detection. While image processing and machine learning techniques have been successfully used to detect plant diseases, crop maturity, and mushroom harvest readiness, these systems often analyze crops at a general level rather than providing detailed monitoring inside the mushroom growing environment. As a result, farmers still face difficulties in accurately tracking the growth stages of mushrooms across multiple racks in a mushroom house. A review of existing research and agricultural monitoring tools highlights several important limitations.

1. Lack of Rack-Wise Monitoring Systems

- Most existing mushroom monitoring systems analyze mushroom images to detect maturity or harvesting readiness, but do not support rack-level monitoring.
- In real mushroom farms, mushrooms grow on multiple racks, and each rack may contain mushrooms at different growth stages.
- Without rack-wise monitoring, farmers cannot easily identify which racks are ready for harvesting.

2. Limited Biological Stage Detection

- Many existing systems only detect whether mushrooms are ready or not ready for harvest.
- They do not clearly identify different biological growth stages, such as early growth, pinning stage, mature stage, and ready-to-harvest stage.
- Because of this limitation, farmers may not fully understand the growth progress of mushrooms.

3. Lack of Integration with Mobile Monitoring Applications

- Some research systems require specialized equipment or complex computer systems, which may not be practical for small-scale farmers.
- Many solutions also lack simple mobile applications that allow farmers to easily capture images and monitor mushroom growth in real time.

4. Limited Support for Harvest Time Prediction

- Although some systems can detect mushroom maturity, they often do not estimate the remaining days until harvesting.
- Without harvest prediction, farmers may harvest mushrooms too early or too late, which affects product quality and yield.

Comparison Between Existing Systems and Proposed System

Feature / Aspect	Existing Systems / Research	Limitations	Proposed System	Reference / Proof
Image-Based Stage Detection	Several studies use image analysis or deep learning (CNN) to detect mushroom maturity or harvest readiness.	Only general maturity detection; do not identify detailed biological stages.	Detects **early growth, pinning, mature, and ready-to-harvest stages** automatically from images.	Husain et al., 2024
Rack-Wise Monitoring	No existing systems provide structured monitoring of individual racks in mushroom farms.	Difficult for farmers to track mushrooms at different stages on multiple **No table of figures entries found.**racks.	Allows farmers to **track growth stages separately for each rack** via a mobile app.	Surige et al., 2021
Harvest Time Estimation	Some systems estimate maturity or readiness based on image features.	Estimates are basic; do not provide accurate **days until harvest**.	Calculates **remaining days for harvesting** based on detected stage.	Moysiadis et al., 2021
Integration with Farm Management	Some smart systems combine sensors and CNN models for environmental monitoring.	Focused on environmental parameters or disease detection; lack **actionable farm management** features.	Integrates **rack-wise stage tracking + harvest timing + mobile app** for practical farm management.	Rahmawati et al., 2018
Ease of Use / Mobile Access	Research experiments often require desktops or lab environments.	Not accessible to farmers in real mushroom farms.	User-friendly **mobile application** for field use, allowing **image capture and rack selection on-site**.	Husain et al., 2024

Table 1Comparison of research gap

2. OBJECTIVE

2.1 Main Objective

Design and implement a Biological Stage Detection Component by October 2026 that accurately identifies the growth stage of oyster mushrooms (early growth, pinning, mature, ready-to-harvest) from images with at least 90% accuracy, and estimates the remaining days to harvest, displaying this information clearly in the app for the farmer.

2.2 Specific Objectives

- Image Capture & Upload: Implement a camera interface on the farmer's portable device by May 2026 that captures mushroom rack images and uploads them to the system in less than 3 seconds per image.
- Stage Detection Model: Develop a machine-learning model by July 2026 that classifies oyster mushroom growth stages (early growth, pinning, mature, ready-to-harvest) with at least 90% accuracy on test datasets.
- Harvest Estimation: Integrate an algorithm to estimate remaining days to harvest by August 2026, aiming for an error margin of ±1 day compared to expert farmer predictions.
- Visual Display & Alerts: Design an app interface by September 2026 that clearly shows detected growth stages and harvest timing, including color-coded stage indicators and optional notifications for harvest readiness.
- Usability & Farmer Effectiveness: Conduct user testing with at least 10 farmers by September 2026, aiming for ≥ 85% of users reporting that the component helps them monitor mushroom growth more efficiently than manual observation.

3. METHODOLOGY

The Software Development Life Cycle (SDLC) is a methodology for designing, developing, testing, and deploying software. It provides a systematic approach to ensure software is developed efficiently and effectively. For this project, an Agile software development approach is applied, emphasizing flexibility, iterative improvement, and collaboration with farmers to gather feedback during development. Short development cycles, or sprints, allow continuous refinement of the Biological Stage Detection Component based on testing results and user input.



Figure 1. 3.01

1. Overall System Description

The Biological Stage Detection Component is implemented as a modular service within the broader mushroom farm management system. Key layers:

- Frontend (Mobile App): Built with React Native to support both Android and iOS devices. The app provides a camera interface for image capture, a dashboard displaying detected mushroom stages, estimated days to harvest, and optional notifications for harvest readiness.
- Backend (Server): Implemented with Node.js + Express.js, which handles image uploads, communicates with the machine learning model, and returns stage detection and harvest predictions to the app.
- Database: MongoDB stores uploaded images, detected growth stages, harvest estimation data, and user information. MongoDB's flexible schema allows easy storage of image metadata, predictions, and historical growth data. Real-time updates can be

implemented via change streams or backend APIs to ensure farmers see current information instantly.

- Machine Learning Layer: A Convolutional Neural Network (CNN) is trained using labeled mushroom images to detect growth stages (early growth, pinning, mature, ready-to-harvest). Harvest estimation is performed via a growth-rate algorithm calibrated with expert farmer data. The model can be hosted on a cloud or edge device for fast inference.
- Integration Points: Detected stage data and harvest timing are displayed directly in the app interface. Optional notifications alert farmers when mushrooms are approaching readiness for harvest, ensuring timely action.
- Validation and Performance Metrics: Stage detection accuracy is targeted at ≥90%, harvest estimation within ±1 day of expert predictions, and at least 85% of users report improved efficiency compared to manual monitoring.

4. HIGH-LEVEL SYSTEM ARCHITECTURE

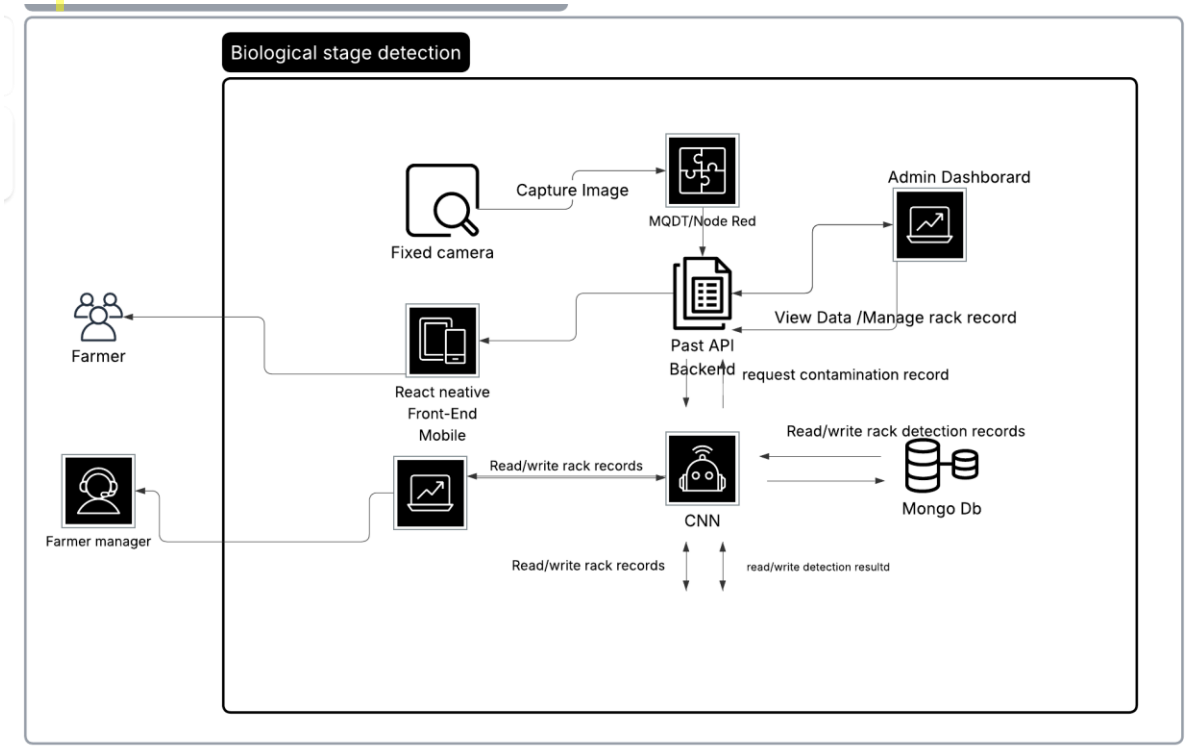


Figure 2 . 4..01 High Level Diagram

4.1 Implementation Steps

The Biological Stage Detection Component is implemented through several structured steps to ensure accurate identification of oyster mushroom growth stages and reliable harvest predictions. The process integrates image acquisition, machine learning analysis, and user-friendly output for farmers.

Step 1: Image Capture and Data Collection

The first step involves capturing images of oyster mushrooms growing on racks. A camera attached to a portable device takes clear photographs of the mushroom clusters at regular intervals. These images serve as the primary data source for the system. The images are collected under controlled lighting conditions to ensure consistency and accuracy in analysis.

Step 2: Image Preprocessing

After capturing the images, preprocessing techniques are applied to improve image quality and prepare them for analysis. This includes resizing images, adjusting brightness and contrast, and

removing noise. Image preprocessing helps standardize the dataset and improves the accuracy of the detection model

Step 3: Feature Extraction and Image Processing

The processed images are analyzed using image processing techniques to extract important visual features of the mushrooms. These features include mushroom size, shape, color, and cap development. The extracted features represent key biological indicators used to determine the growth stage of the mushroom.

Step 4: Biological Stage Detection Model

A machine learning model is used to classify the mushroom growth stage based on the extracted features. The model is trained using labeled images of different mushroom growth stages such as early growth, pinning, mature, and ready-to-harvest stages. The trained model predicts the current biological stage of the mushroom accurately.

Step 5: Harvest Time Estimation

Once the growth stage is identified, the system estimates the remaining time required for harvesting. This estimation is based on growth patterns observed in the training data and environmental conditions. The system calculates the approximate number of days left before the mushrooms reach the optimal harvesting stage.

Step 6: Information Display and System Integration

Finally, the detected biological stage and estimated harvest time are displayed in the mobile application used by the farmer. The information is stored in the system database and presented in a simple and clear format. This allows farmers to monitor mushroom growth easily and make timely harvesting decisions.

5. USER REQUIREMENTS

5.1 Stakeholder Identification

The key stakeholders for the Biological Stage Detection Component include farmers, owners, system administrators, and agricultural experts. Farmers are the primary users who need to monitor the growth stages of oyster mushrooms and track harvest timing efficiently. The owner oversees multiple mushroom racks and requires summarized information and alerts to plan harvesting effectively. System administrators are responsible for maintaining the system, updating the machine learning model, and ensuring overall reliability and security. Finally, agricultural experts provide labeled data for model training and help validate the accuracy of the growth stages detected**.**

Stakeholder	Role
Farmers	Primary users need to monitor mushroom growth stages and harvest timing efficiently.
Owner	Oversees multiple racks; wants summaries and alerts for harvest planning.
System Administrator	Maintains the system, updates the machine learning model, and ensures reliability and security.
Agricultural Experts	Provide labelled data and validate growth stage accuracy.

Table 2 Stakeholer of the system

5.2 Functional Requirements

FR No.	Input	Process	Output
FR-01	Image of mushroom rack (JPG, PNG)	System accepts and validates the uploaded image.	Image stored in backend for processing.
FR-02	Uploaded mushroom image	Preprocessing (resizing, noise removal, contrast adjustment) applied.	Cleaned image ready for ML model.
FR-03	Preprocessed mushroom image	CNN-based ML model detects growth stage (early growth, pinning, mature, ready-to-harvest).	Detected growth stage for each mushroom rack.
FR-04	Detected growth stage + size data	Harvest estimation algorithm calculates remaining days to harvest.	Estimated days to harvest (±1 day accuracy).
FR-05	Detected growth stage and harvest estimate	Dashboard module formats data for display.	Visual dashboard with color-coded stages and harvest timing.
FR-06	User preferences for notifications	Notification system triggers alerts when mushrooms approach harvest.	Push notifications or in-app alerts for harvest readiness.
FR-07	User corrections/feedback	System updates ML model training dataset.	Improved accuracy over time via model retraining.
FR-08	Historical images and stage data	History tracking stores all past records in MongoDB.	Historical log of images, growth stages, and harvest estimates.
FR-09	User login credentials	Authentication system verifies identity.	Secure access to user-specific mushroom data.
FR-10	User query (filter by rack/date)	Filtering algorithm retrieves matching records.	Filtered list of mushroom stage and harvest data by criteria.

Table 3 Funtional requirement

5.3 Non-Functional Requirements

NFR No.	Input	Process	Output
NFR-01	User request to upload image	Backend ensures image is validated, sanitized, and securely transmitted.	Secure and safe image storage without corruption or unauthorized access.
NFR-02	System requests stage detection	ML model processes image efficiently.	Stage detection results returned within 3 seconds per image.
NFR-03	Harvest estimation request	Algorithm calculates remaining days based on growth stage and size.	Harvest estimation within ±1 day of expert farmer prediction.
NFR-04	User accessing dashboard	Frontend renders UI on mobile devices (Android/iOS).	Intuitive, responsive, and user-friendly interface.
NFR-05	Multiple users using system simultaneously	Backend handles concurrent requests via Node.js + Express + MongoDB.	Reliable system performance with no crashes or slowdowns; ≥99% uptime.
NFR-06	System storing large historical data	Database (MongoDB) stores and indexes images and metadata.	Scalable data storage capable of handling growing farms and racks.
NFR-07	ML model updates or retraining	Developer updates model and deploys to server/cloud.	Maintainable system where improvements do not disrupt service.
NFR-08	Data access by authorized users	Authentication and role checks in backend.	Secure access control ensuring only authorized farmers view data.
NFR-09	System under heavy load or network delay	Backend manages requests with asynchronous processing and retries.	Stable and consistent system performance under stress.
NFR-10	Farmer requests notifications	Notification service checks thresholds and triggers alerts.	Timely push notifications or in-app alerts for harvest readiness.

Table 4 Non Funtional requirement

6. COMMERCIALIZATION PLAN

I. Target Market and Customer Persona

The primary target market for this system is newly started mushroom farmers and small-scale oyster mushroom farm owners. Many new farmers lack experience in identifying the correct mushroom growth stages and determining the right harvesting time. Because of this, they may harvest mushrooms too early or too late, which can reduce product quality and profit.

A typical customer persona is a young entrepreneur or beginner farmer aged between 20–40 years who has recently started a mushroom cultivation business with a small number of growing racks. These farmers usually depend on manual observation and basic knowledge to monitor mushroom growth. Therefore, they require a simple, affordable, and easy-to-use technological solution that helps them understand mushroom growth stages and plan harvesting more efficiently.

II. Cost Estimation

The major costs for this system include software development, machine learning model training, sensors for environmental monitoring, cloud hosting services, database management, and system maintenance. Additional expenses may include system upgrades, technical support, and farmer training sessions.

III. Competitive Advantage

This system provides several advantages compared to traditional manual monitoring methods. It offers accurate mushroom growth stage detection using machine learning, automated harvest time estimation, and a simple mobile application interface. The system is specifically designed for beginner farmers, making it easy to use even for users with limited technical knowledge.

7. BUDGET AND JUSTIFICATION

The budget for this project includes the resources required to design, develop, test, and deploy the Biological Stage Detection System for oyster mushrooms. The costs mainly cover hardware used for development, sensors required for environmental monitoring, software tools, cloud services for hosting the backend and machine learning model, and human effort required during development. Each expense is justified based on its importance for building and demonstrating the system effectively.

Item	Description	Cost (LKR)
Hardware Costs	Laptops and mobile devices are used for coding, machine learning model training, system development, and testing the mobile application.	50,000
Sensors	Environmental sensors (temperature and humidity sensors) used to monitor mushroom growing conditions and support the biological stage detection process.	10,000
Software Licenses	Development tools such as IDEs, database tools, and supporting libraries required for system development and testing.	10,000
Cloud Services	Cloud hosting for the backend server, machine learning inference API, and MongoDB database used to store images and prediction data.	20,000
Human Effort Cost	Time and effort spent on system design, coding, machine learning model training, testing, and debugging during development.	15,000
Internet & Utilities	Internet connectivity and electricity required for development, testing, and uploading data to the cloud server.	3,000
Contingency	Reserved fund to handle unexpected expenses during project development or testing.	2,000
Total		110,000

Table 5 Budget

8. WORK BREAKDOWN STRUCTURE (WBS)

The Work Breakdown Structure (WBS) provides a detailed hierarchical breakdown of all tasks required to develop the Biological Stage Detection System for oyster mushrooms. It helps in planning, managing, and monitoring the project efficiently by dividing it into smaller, manageable tasks with clear responsibilities, timelines, and workload distribution.

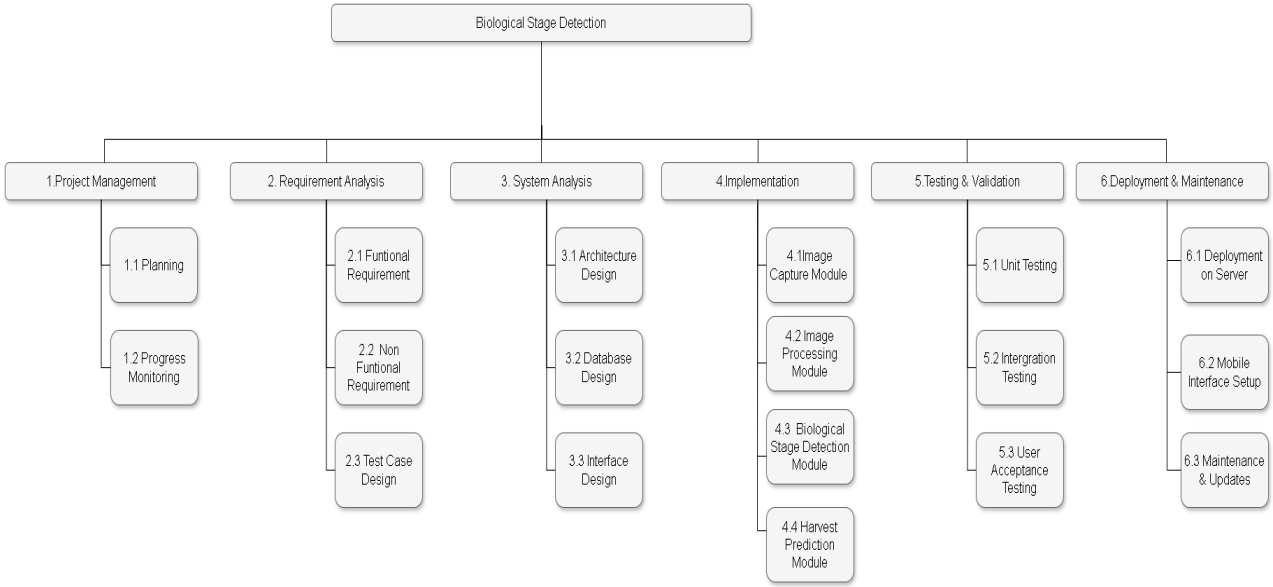
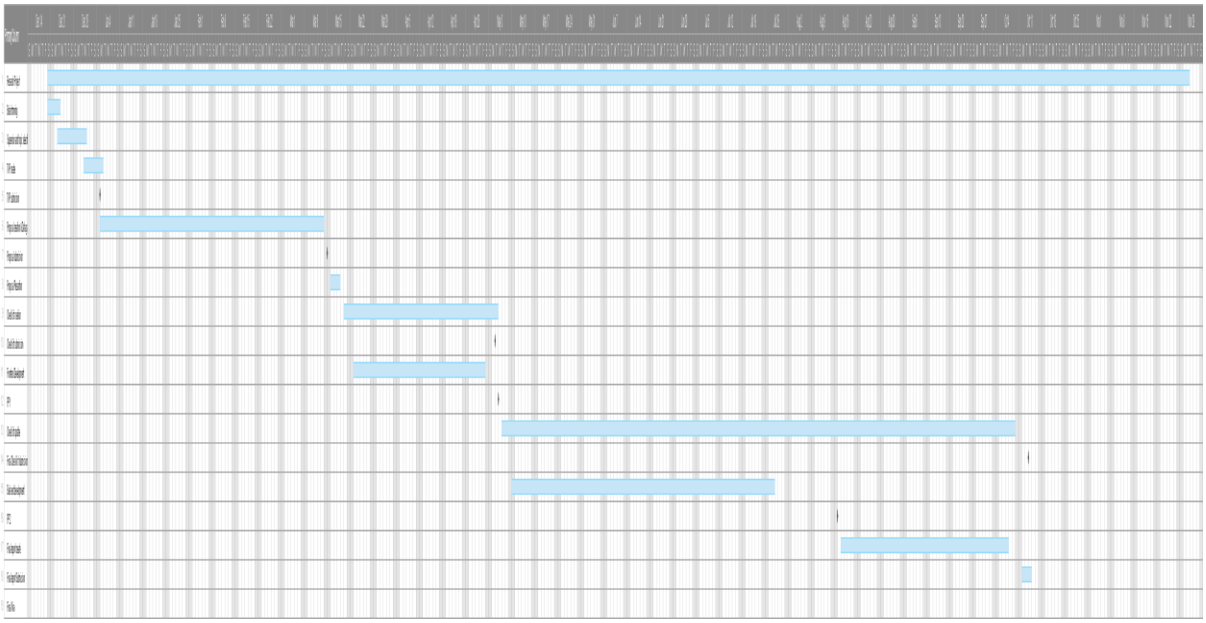


Figure 3.8.01. WBS

9. GANTT CHART



https://app.smartsheet.com/workspaces/fxHV3XjX269HFMm9j73qRxXvJChcpgppvHV3hXQ1 (Gantt Chart)

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11.Appendix

1. https://app.smartsheet.com/workspaces/fxHV3XjX269HFMm9j73qRxXvJChcpgppvHV3hXQ1 (Gantt Chart)

2. Observation Checklist

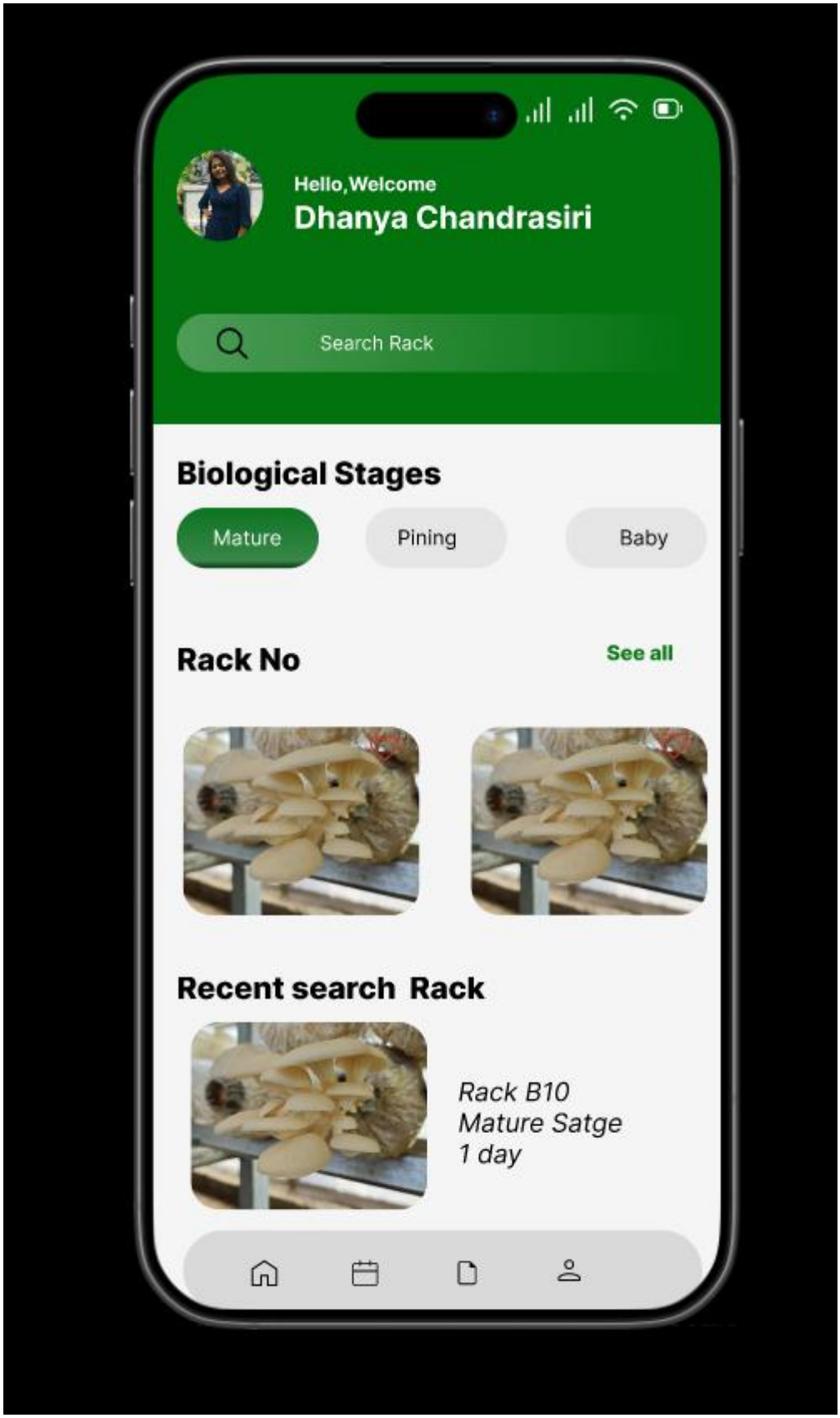
Observation Checklist

Farm Name: _____

Location: _____ Date: _____

Area to Observe	Yes	No	Comments
Environment			
Temperature within optimal range (24–28°C)	☐	☐	
Humidity level maintained (80–95%)	☐	☐	
Ventilation adequate	☐	☐	
Lighting schedule followed as planned	☐	☐	
Growing Conditions			
Moisture level of substrate maintained	☐	☐	
Mushroom racks properly organized	☐	☐	
Fresh air exchange system functional	☐	☐	
Crop Health			
Healthy mushroom growth observed	☐	☐	
Discoloration/deformities	☐	☐	
Presence of pests or insects	☐	☐	
Contamination Checks			
Signs of mold or contamination in bags	☐	☐	
Hygiene practices observed by workers	☐	☐	
Operational Practices			
Monitoring equipment functional	☐	☐	
Record-keeping updated regularly	☐	☐	

3. UI


Hello,Welcome
Dhanya Chandrasiri
Search Rack
Biological Stages
Mature
Pining
Baby
Rack No
See all
Recent search Rack
Rack B10
Mature Satge
1 day


User name
Log in